

CV/Gate Distribution Panel

Medium-sized and large music synthesizers often require external control voltage and gate signals. This simple circuit, complete with a custom front panel design, makes effective use of the unused space beneath the "Power" front panel.

The CV and gate outputs from the interface receiver—which are hardwired to the VCOs and VCFs or ADSRs—are connected here to op-amp voltage followers (buffers); their outputs are then distributed to six 3.5 mm jack sockets each (Figure 1). Capacitors C1 and C2 serve to suppress interference.

If you wish to make the circuit board even more compact, you can use a dual op-amp instead of two single op-amps. The MC or LM 1458 IC (from Motorola or National Semiconductor, respectively)—the pinout of which is shown in Figure 3—is functionally identical to the well-known 741. Figure 4 shows a suggested front panel design. The dimensions of the front panel correspond to half of a "large" FORMANT front panel.

Applications

This output splitter has proven particularly useful in conjunction with the voltage control of LFOs and waveform processors (Chapter 5), as well as VC phasers and other voltage-controlled modules.

Furthermore, the KB Gate output can be used to synchronize the keyboard and sequencers. The KOV/KB Gate module has also proven effective when using multiple module enclosures, as no additional interface receivers are required.

Finally, here is a tip: if you connect a KOV output to the FM input of a VCO (with the KOV input disabled), the intervals between the keys can be reduced as desired. This makes quarter-tone and eighth-tone steps possible. The wiring scheme is shown in Figure 5; any waveform can be used.

Parts list for Figure 1

Capacitors (MKH, MKM):

C1, C2 = 100 nF

Semiconductors:

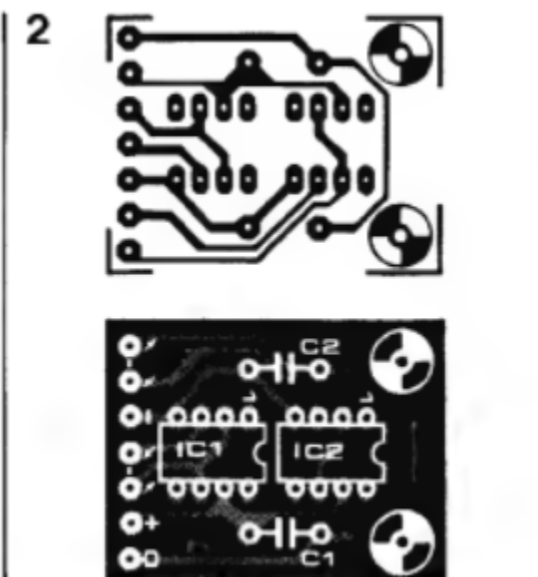
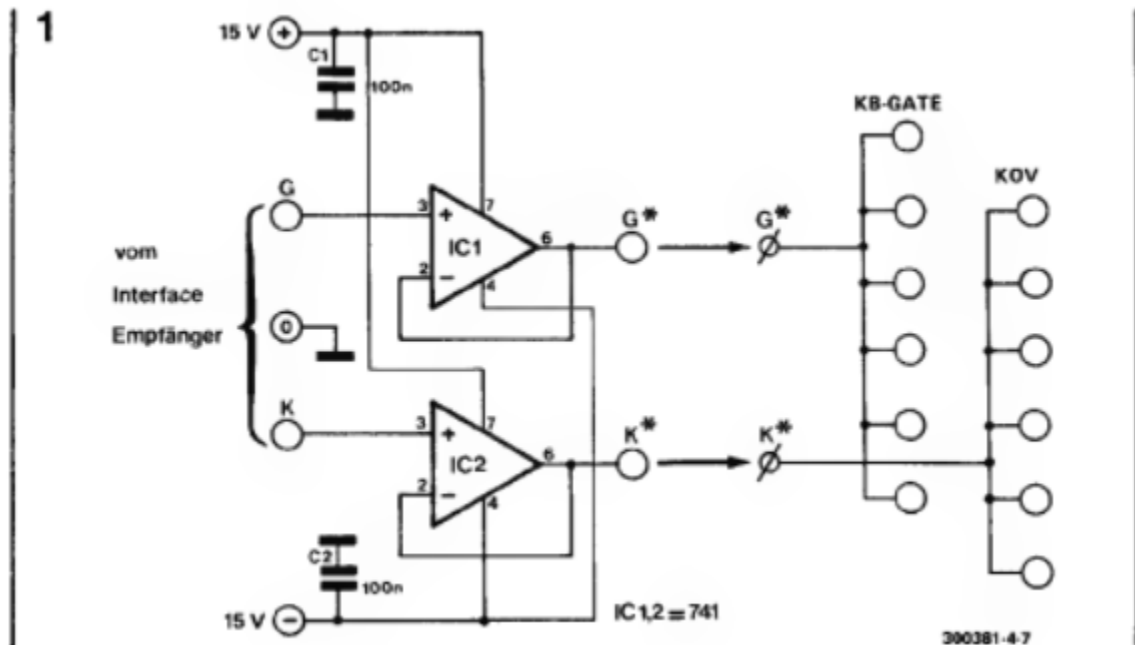
IC1, IC2 = 741 (1 x MC 1458, LM 1458)

Miscellaneous:

12 x 3.5 mm jack sockets

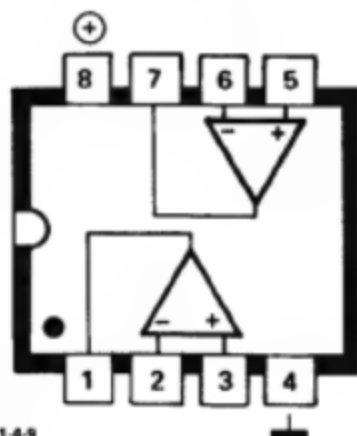
Figures

- Figure 1. Connection diagram for the KOV/KB gate distribution panel.
- Figure 2. PCB layout and component placement plan for the distribution panel.
- Figure 3. Pinout of the LM 1458 IC.
- Figure 4. Proposed front panel design for KOV and KB gate.
- Figure 5. Wiring diagram for reducing the intervals between keys.



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LM 1458 N



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KOV**OUT****KB-
GATE****OUT**